

COAL LAB-02

Task#01(a):

The screenshot shows the Microsoft Visual Studio IDE with the following components:

- Code Editor:** Contains the assembly code for `task1.asm`. The code implements the equation $a) 0110010 - 083h + 285 + 20 + 10001101 - 0E + 0F$. The code includes Irvine32.inc, defines a main PROC, and uses instructions like `mov`, `sub`, `add`, `call`, and `exit`.
- Output Window:** Displays the build output. It shows the project configuration (Lab-02, Debug Win32) and the assembly process. The build succeeded, and the executable `Lab-02.exe` was generated.
- Debugger:** The Local Windows Debugger is active, showing the execution of the program. The register values are displayed in the register window.
- Register Window:** Shows the state of the registers after execution. The values are: EAX=FFFFFF4E, EBX=01088000, ECX=00E410FF, EDX=00E410FF, ESI=00E410FF, EDI=00E410FF, EBP=012FF804, ESP=012FF7F8, EIP=00E43682, EFL=00000292, CF=0, SF=1, ZF=0, OF=0, AF=1, PF=0.
- Task List:** Shows the task list with the task `task1.asm` listed.

Task#01(b):

The screenshot shows the Microsoft Visual Studio IDE with the following components:

- Code Editor:** Contains the assembly code for `task2.asm`. The code implements the equation $b) 45 + 69 + 50 + 75 + 54 + 440 - 0Ah$. The code includes Irvine32.inc, defines a main PROC, and uses instructions like `mov`, `add`, `sub`, `call`, and `exit`.
- Output Window:** Displays the build output. It shows the project configuration (Lab-02, Debug Win32) and the assembly process. The build succeeded, and the executable `Lab-02.exe` was generated.
- Debugger:** The Local Windows Debugger is active, showing the execution of the program. The register values are displayed in the register window.
- Register Window:** Shows the state of the registers after execution. The values are: EAX=0000013F, EBX=00E28000, ECX=00B210FF, EDX=00B210FF, ESI=00B210FF, EDI=00B210FF, EBP=010FFB10, ESP=010FFB04, EIP=00B2367C, EFL=00000216, CF=0, SF=0, ZF=0, OF=0, AF=1, PF=1.
- Task List:** Shows the task list with the tasks `task1.asm`, `task2.asm`, and `task2.asm` listed.

Task#02:

Registers
EAX = 00000005 EBX = 00000009 ECX = 000110FF EDX = 000110FF ESI = 000110FF EDI = 000110FF EIP = 0001366A ESP = 000EFC5C EBP = 000EFC68 EFL = 00000246

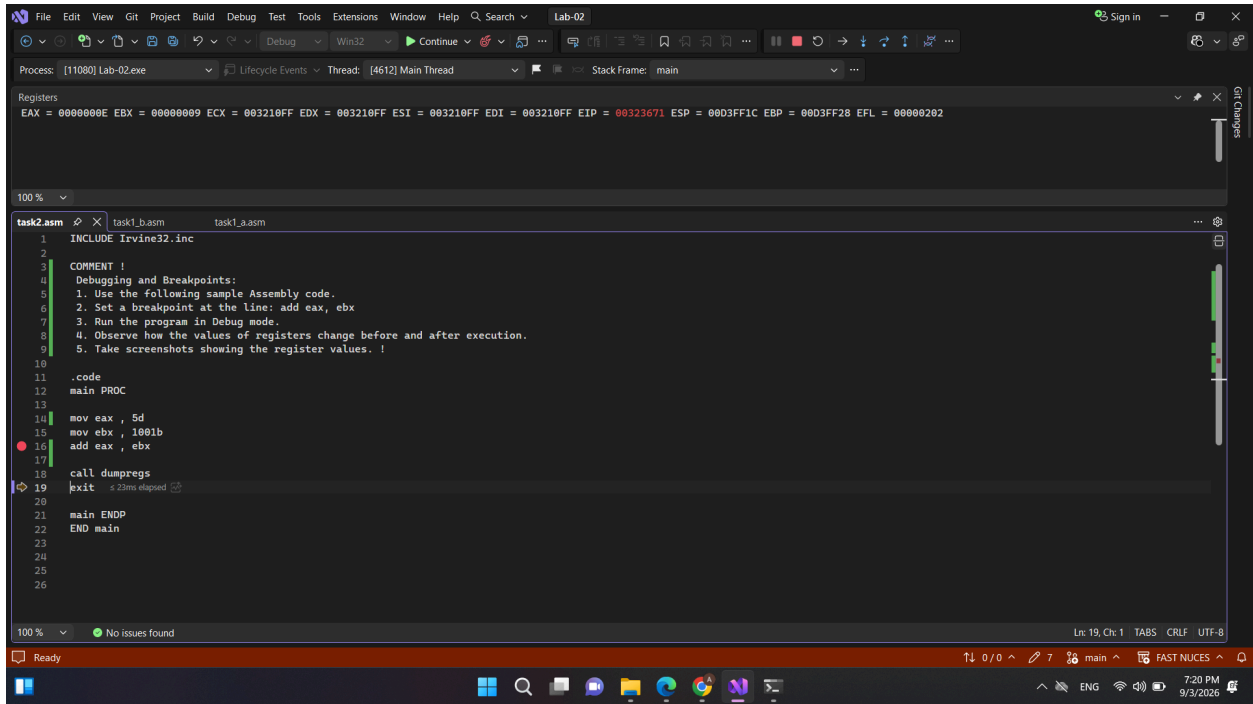
```
1 INCLUDE Irvine32.inc
2
3 COMMENT !
4 Debugging and Breakpoints:
5 1. Use the following sample Assembly code.
6 2. Set a breakpoint at the line: add eax, ebx
7 3. Run the program in Debug mode.
8 4. Observe how the values of registers change before and after execution.
9 5. Take screenshots showing the register values. !
10
11 .code
12 main PROC
13
14 mov eax, 5d
15 mov ebx, 1001b
16 add eax, ebx
17
18 call dumpregs
19 exit
20
21 main ENDP
22 END main
23
24
25
26
```

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Registers
EAX = 0000000E EBX = 00000009 ECX = 00C210FF EDX = 00C210FF ESI = 00C210FF EDI = 00C210FF EIP = 00C2366C ESP = 010FF7FC EBP = 010FF808 EFL = 00000202

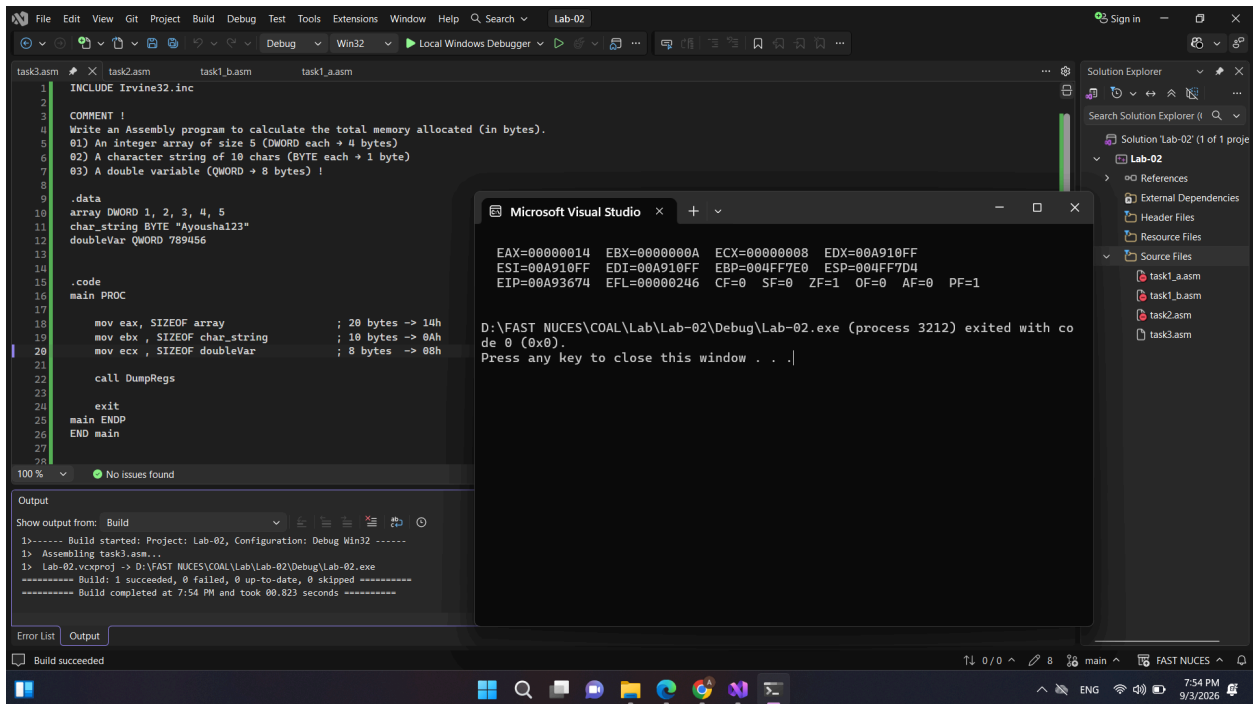
```
1 INCLUDE Irvine32.inc
2
3 COMMENT !
4 Debugging and Breakpoints:
5 1. Use the following sample Assembly code.
6 2. Set a breakpoint at the line: add eax, ebx
7 3. Run the program in Debug mode.
8 4. Observe how the values of registers change before and after execution.
9 5. Take screenshots showing the register values. !
10
11 .code
12 main PROC
13
14 mov eax, 5d
15 mov ebx, 1001b
16 add eax, ebx
17
18 call dumpregs  s 1ms elapsed
19 exit
20
21 main ENDP
22 END main
23
24
25
26
```

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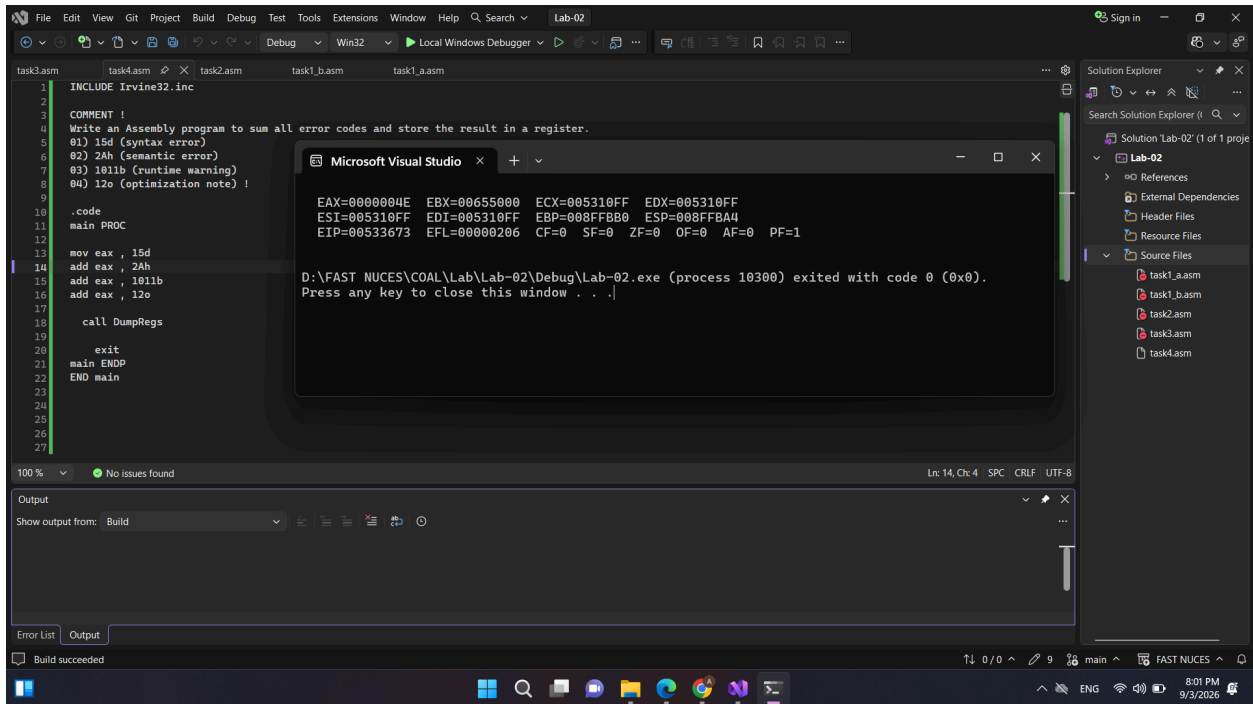


(Debugging end here)

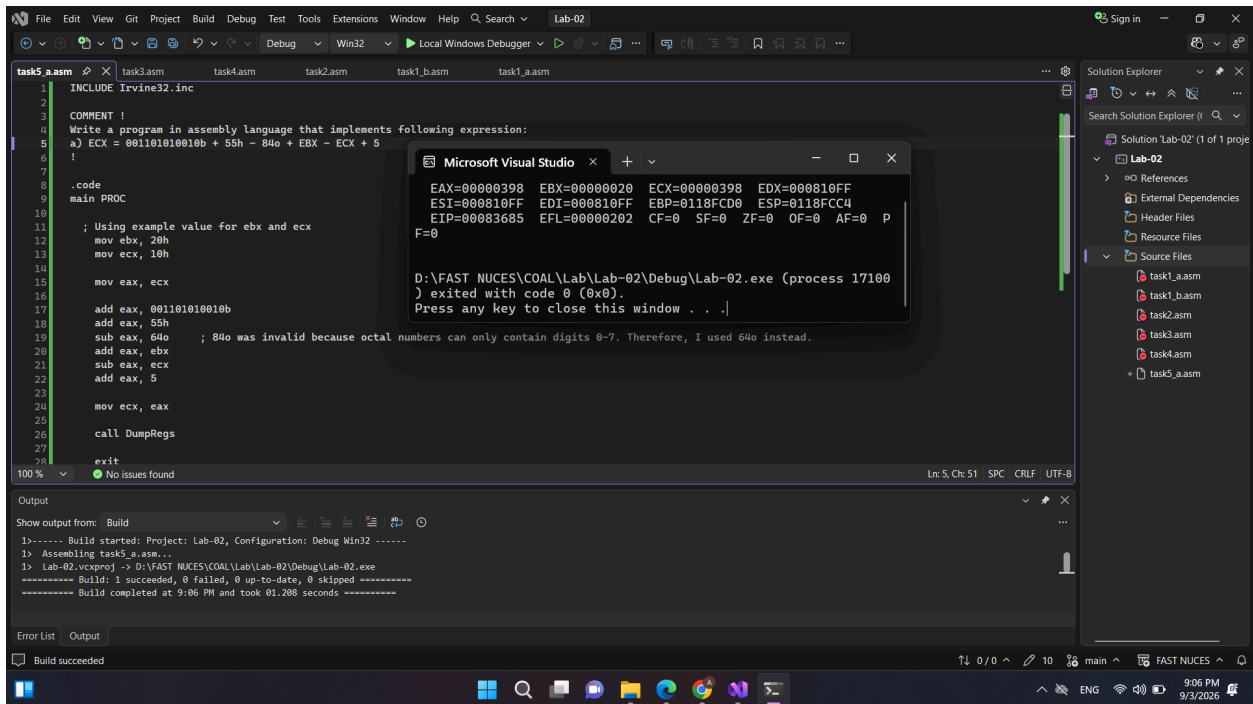
Task#03:



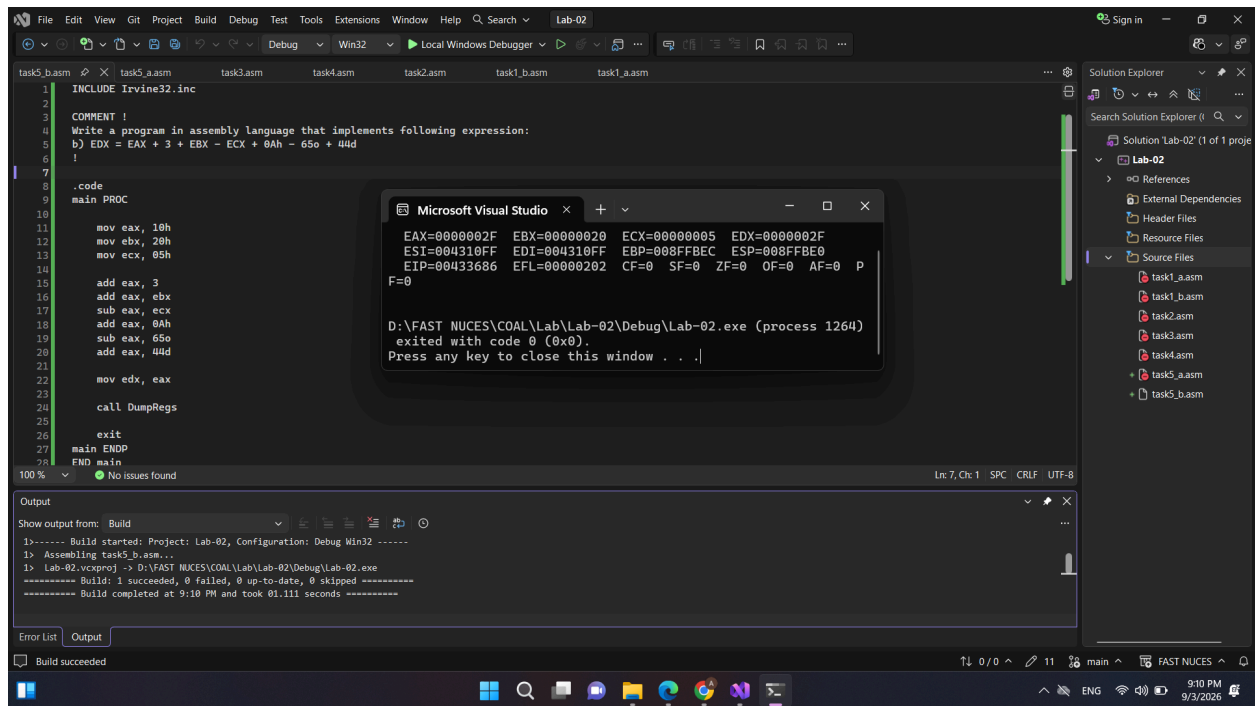
Task#04:



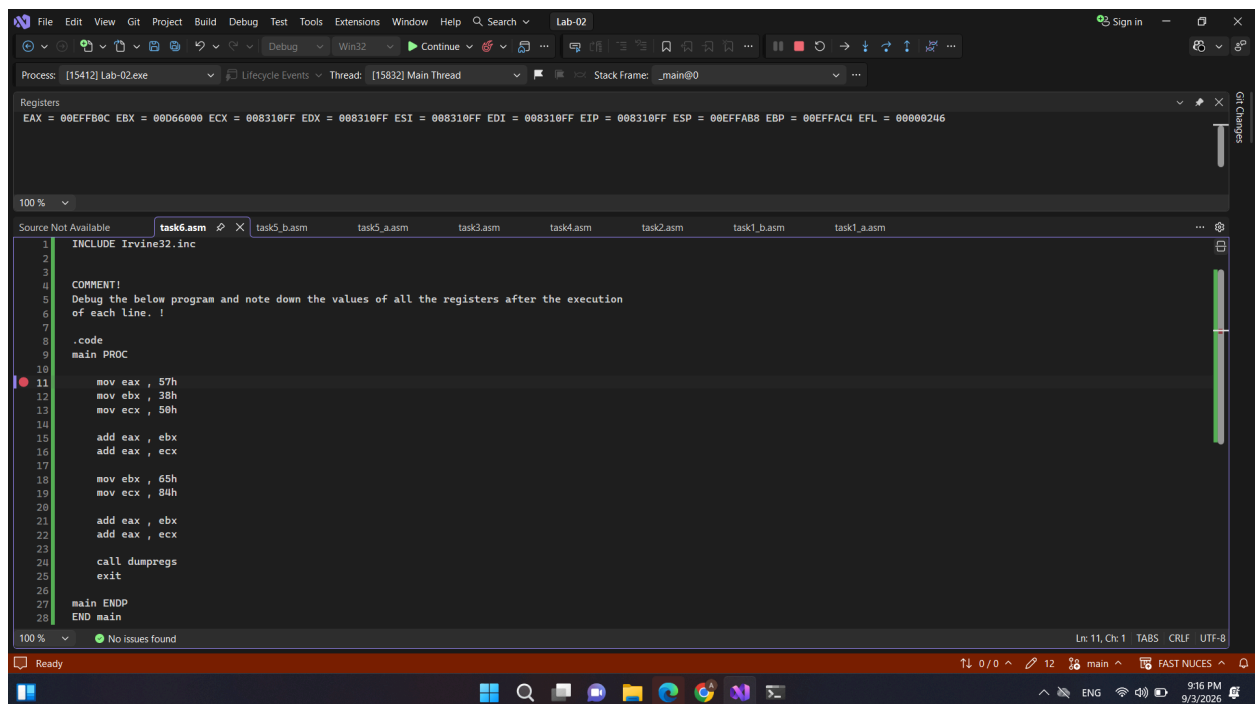
Task#05(a):



Task#05(b):



Task#06: Screenshot after every Fn+F10:



File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab-02

Process: [15412] Lab-02.exe Thread: [15832] Main Thread Stack Frame: main

Registers
EAX = 00EFFB0C EBX = 00D66000 ECX = 008310FF EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 00833660 ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000246

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

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Ready

0/0 12 main FAST NUDES 9:17 PM 9/3/2026

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Process: [15412] Lab-02.exe Thread: [15832] Main Thread Stack Frame: main

Registers
EAX = 00000057 EBX = 00D66000 ECX = 008310FF EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 00833665 ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000246

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

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Ready

0/0 12 main FAST NUDES 9:17 PM 9/3/2026

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Process: [15412] Lab-02.exe Lifecycle Events Thread: Stack Frame:

Registers
EAX = 00000057 EBX = 00000030 ECX = 008310FF EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 0083366A ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000246

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

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Ready 0/0 12 main FAST NUDES 9:17 PM 9/3/2026

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab-02

Process: [15412] Lab-02.exe Lifecycle Events Thread: [15832] Main Thread Stack Frame: main

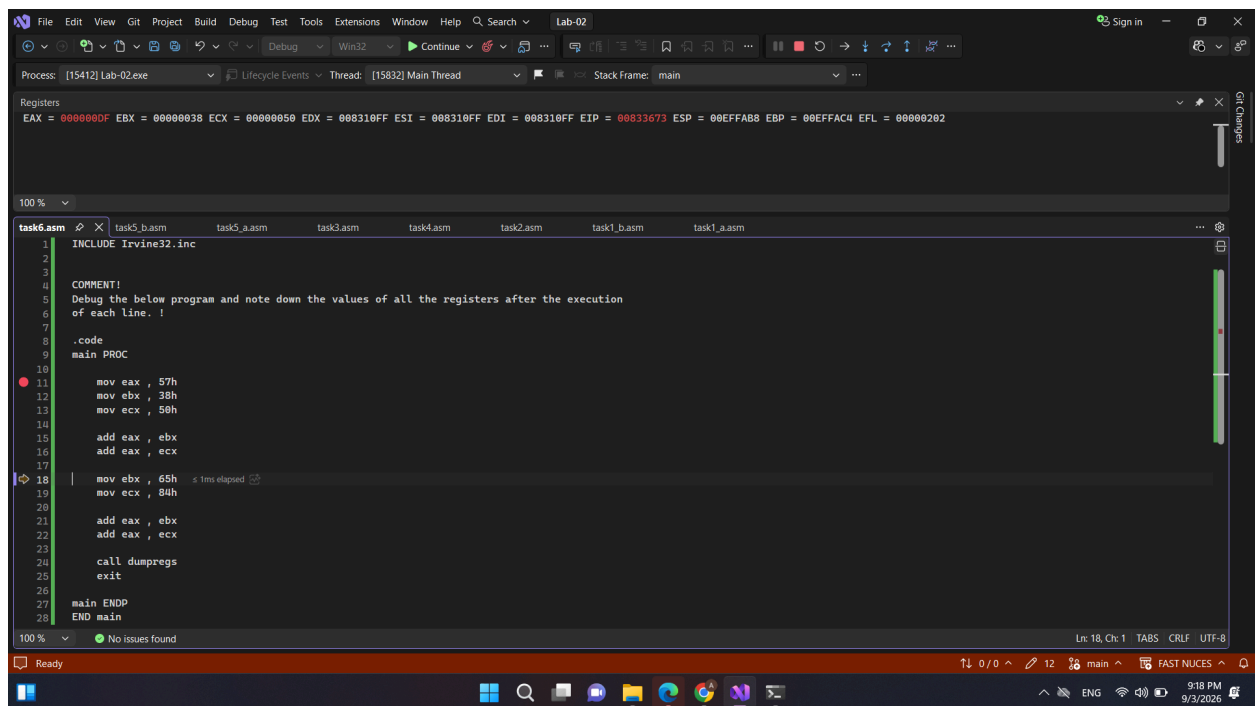
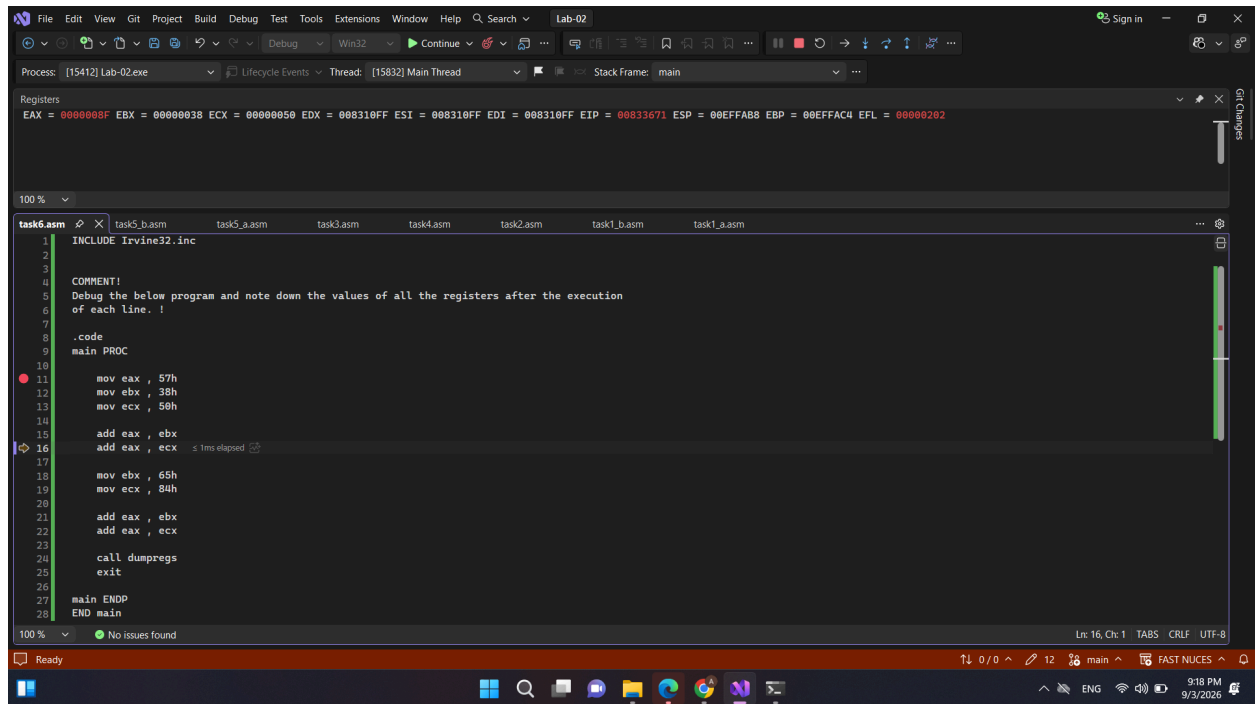
Registers
EAX = 00000057 EBX = 00000038 ECX = 00000050 EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 0083366F ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000246

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

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Process: [15412] Lab-02.exe Lifecycle Events Thread: [15832] Main Thread Stack Frame: main

Registers
EAX = 0000000F EBX = 00000065 ECX = 00000050 EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 00833678 ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000202

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

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Process: [15412] Lab-02.exe Lifecycle Events Thread: [15832] Main Thread Stack Frame: main

Registers
EAX = 0000000F EBX = 00000065 ECX = 00000084 EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 0083367D ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000202

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

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Ready 0/0 12 main FAST NUDES 9:19 PM 9/3/2026

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Process: [15412] Lab-02.exe Lifecycle Events Thread: [15832] Main Thread Stack Frame: main

Registers
EAX = 00000144 EBX = 00000065 ECX = 00000084 EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 0083367F ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000216

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

Ln: 22, Ch: 1 TABS CRLF UTF-8

Ready 0/0 12 main FAST NUDES 9:19 PM 9/3/2026

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search Lab-02

Process: [15412] Lab-02.exe Lifecycle Events Thread: [15832] Main Thread Stack Frame: main

Registers
EAX = 000001C8 EBX = 00000065 ECX = 00000084 EDX = 008310FF ESI = 008310FF EDI = 008310FF EIP = 00833681 ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000202

task6.asm task5_b.asm task5_a.asm task3.asm task4.asm task2.asm task1_b.asm task1_a.asm

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

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Ready 0/0 12 main FAST NUDES 9:19 PM 9/3/2026

Registers

EAX = 000001C8 EBX = 00000065 ECX = 00000084 EDX = 000310FF ESI = 000310FF EDI = 000310FF EIP = 00033686 ESP = 00EFFAB8 EBP = 00EFFAC4 EFL = 00000202

```
1 INCLUDE Irvine32.inc
2
3
4 COMMENT!
5 Debug the below program and note down the values of all the registers after the execution
6 of each line. !
7
8 .code
9 main PROC
10
11 mov eax, 57h
12 mov ebx, 38h
13 mov ecx, 50h
14
15 add eax, ebx
16 add eax, ecx
17
18 mov ebx, 65h
19 mov ecx, 84h
20
21 add eax, ebx
22 add eax, ecx
23
24 call dumpregs
25 exit
26
27 main ENDP
28 END main
```

Task#07:

Today's lab helped me understand Assembly Language debugging using breakpoints and single stepping. I learned how to observe register values and trace instructions during execution. I also learned about identifiers, instructions, directives, comments, and data types such as BYTE, WORD, DWORD, and QWORD. In addition, I learned about signed and unsigned data types and different number systems. Overall, the lab improved my understanding of registers, memory, and basic Assembly programming.